

Sustainability aware asset management

Lecture 5: Investment Project

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Objectives of the Lecture

- How to calculate inputs?
- How to solve the problem with restrictions?
- How to calculate the allocation in real time?
- How to estimate key characteristics of an allocation experiment?
- How to calculate the ex-post performance of a portfolio?

Objectives of the lecture

- **Financial performance**
- Issues with Missing Values
- Design of an Investment Strategy
- Risk Measures

Simple and Log Return

Holding the asset for one period from t to $t + 1$ yields the **one-period simple return**:

$$R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

where

- P_t : price of the asset at date t ,
- R_{t+1} : one-period simple return from date t to date $t + 1$.

The **continuously compounded (or log-) return** r_{t+1} is

$$r_{t+1} = \log(1 + R_{t+1}) = \log(P_{t+1}) - \log(P_t) = p_{t+1} - p_t$$

where $p_t = \log(P_t)$ is the log-price

Dividend Payments and Excess Returns

For assets with periodic dividend payments, asset returns must be redefined:

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t} - 1$$

and

$$r_{t+1} = \log(P_{t+1} + D_{t+1}) - \log(P_t)$$

where D_{t+1} is the dividend payment of an asset between dates t and $t + 1$ and P_{t+1} is the price of the asset at the end of $t + 1$ (dividend not included in P_{t+1}).

Excess return is the difference between the asset return and the risk-free rate $R_{f,t}$:

$$R_{t+1}^e = R_{t+1} - R_{f,t}$$

and

$$r_{t+1}^e = r_{t+1} - r_{f,t},$$

with $R_{f,t}$ and $r_{f,t}$ the simple return and log return of the risk-free asset. We use the index t because the risk-free rate between t and $t + 1$ is determined in t .

Definition of Volatility

The unconditional volatility is estimated as the sample standard deviation:

$$s = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2} \quad \text{or} \quad \hat{\sigma} = \sqrt{\frac{1}{T} \sum_{t=1}^T (R_t - \bar{R})^2}$$

where R_t is the simple return on day t and $\bar{R}$ is the sample mean over the T days.

- s is the unbiased estimator
- $\hat{\sigma}$ is the ML estimator, asymptotically unbiased

For the moment, we focus on a unique estimator. We will discuss the case of a dynamic volatility below.

Volatility of a Portfolio

The volatility of a portfolio p can be computed in two ways:

Ex-ante volatility

At the end of period t , we have measured the covariance matrix of returns for $t + 1$, Σ_t , computed over a number of periods before $t + 1$ and determined the weights α_t . Then, we compute the volatility for the next period as

$$\sigma_{p,t} = \sqrt{\alpha_t' \Sigma_t \alpha_t}.$$

Ex-post volatility

At the end of the investment period (over T periods), we have computed the portfolio return $R_{p,t}$ for all dates of the sample. Then, the volatility of the strategy over the T periods is

$$\sigma_p = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_{p,t} - \bar{R}_p)^2},$$

where $\bar{R}_p$ is the average return of the portfolio over the T periods.

Risk-Adjusted Return

The Sharpe ratio measures the return of a strategy per unit of risk (i.e., the volatility):

$$SR_p = \frac{\bar{R}_p - R_f}{\sigma_p},$$

The **Sharpe ratio** is usually computed ex post, at the end of the investment period, to assess the performance of the strategy.

The **Treynor ratio** measures the return of a strategy per unit of systematic risk (i.e., the beta to the market portfolio):

$$TR_p = \frac{\bar{R}_p - R_f}{\beta_p}.$$

Jensen's alpha relies on the same logic of the performance of the portfolio relative to the market portfolio.

The market model is written as

$$R_{p,t+1} - R_{f,t} = \alpha_p + \beta_p(R_{m,t+1} - R_{f,t}) + \varepsilon_{t+1},$$

where

- $R_{m,t}$ is the market portfolio return,
- β_p is the systematic exposure of the portfolio to market risk,
- α_p is the extra performance if positive (or loss if negative) relative to the portfolio performance implied by the market model.

Maximum Drawdown

The maximum drawdown (MDD) characterizes the greatest peak-to-trough loss within a specified temporal span.

Let $\{p_t\}_{t=1}^T$ be a sample path associated with the stochastic process, where p_t denotes the log-price on day t . The drawdown within this period on day t corresponds to the return loss between the last peak and the current price:

$$DD_t = \max_{1 \leq s \leq t} (p_s - p_t).$$

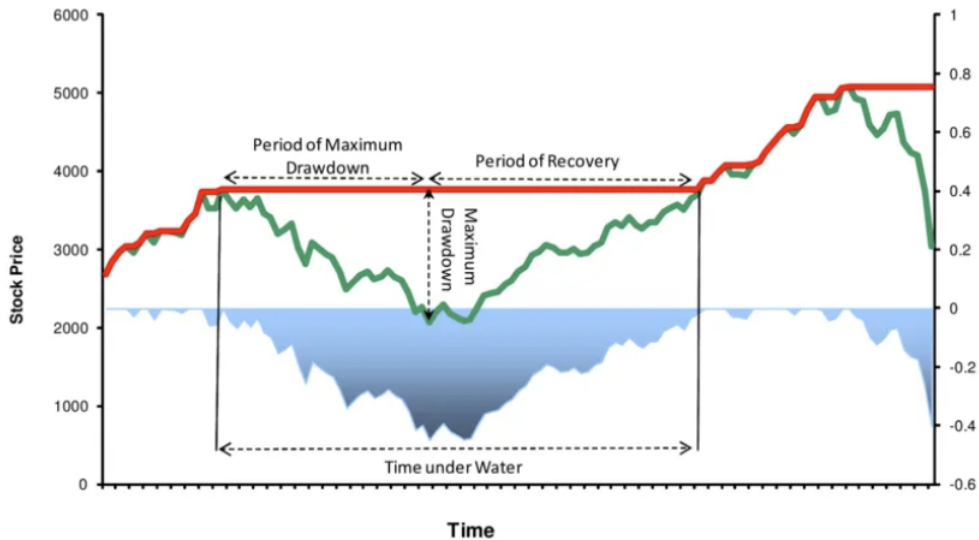
We denote the drawdowns within the sample as $\{DD_t\}_{t=1}^T$.

The maximum drawdown is defined as the largest drawdown of the sample:

$$MDD = \max_{1 \leq t \leq T} DD_t.$$

This measure suffers from limitations: It is an extreme statistic and as such it is highly sensitive to outliers. For this reason, it may not be well-suited for use in an optimization exercise. In addition, it is not a coherent measure of risk (see below).

Maximum Drawdown



Annualization of Performance Measures

Annualization can be used in two contexts:

- Case 1: We have monthly returns and want to express them as annual returns
- Case 2: We have the cumulative performance over T years and want to express it as an annual return

Case 1: From monthly to annual

We have an average monthly return $\bar{R}_p^{(m)}$ with monthly volatility $\sigma_p^{(m)}$. The average annual return and annual volatility are, assuming i.i.d. returns:

$$\bar{R}_p^{(y)} = 12 \bar{R}_p^{(m)} \quad \text{and} \quad \sigma_p^{(y)} = \sqrt{12} \sigma_p^{(m)}.$$

The Sharpe ratio is annualized as:

$$SR_p^{(y)} = \sqrt{12} SR_p^{(m)}.$$

Annualization of Performance Measures

Case 2: From cumulative performance to average annual return

We have cumulated monthly returns over T periods (i.e., $T/12$ years).

- If we use simple returns, the cumulative return is

$$R_p[T] = \prod_{k=1}^T (1 + R_{p,t+k}) - 1.$$

- If we use log returns, the cumulative return is

$$r_p[T] = \sum_{k=1}^T r_{p,t+k},$$

where $R_{p,t+k}$ and $r_{p,t+k}$ denote monthly returns.

The **annualized simple and log returns** over the sample are given by

$$\bar{R}_p^{(y)} = (1 + R_p[T])^{12/T} - 1 \quad \text{and} \quad \bar{r}_p^{(y)} = \frac{12}{T} r_p[T].$$

Turnover

Turnover

is measured as

$$TO_t = \sum_{i=1}^N \left| \alpha_{i,t}^{new} - \alpha_{i,t}^{old} \right|,$$

where $\alpha_{i,t}^{old}$ is the weight of asset i just before the rebalancing.

If the last rebalancing was done at the end of period $t - 1$, we update the weight for the relative change in the value of the securities in the portfolio:

$$\alpha_{i,t}^{old} = \frac{\alpha_{i,t-1}^{new}(1 + R_{i,t})}{\sum_{j=1}^N \alpha_{j,t-1}^{new}(1 + R_{j,t})}.$$

Remark: If we sum the $\alpha_{i,t}^{old}$ terms over all securities, we find

$$\sum_{i=1}^N \alpha_{i,t}^{old} = 1.$$

The Turnover is useful to compute **Transaction cost** TC_t , as we should just multiply by the transaction fee (τ) per dollar bought or sold:

$$TC_t = \tau TO_t.$$

The transaction cost might be different for buying securities, selling securities, or short-selling securities.

Ex-ante Performance Relative to a Benchmark

The vector of weights in the benchmark is denoted by $\alpha_{b,t}$.

The active weights measure the deviation from the benchmark: $\alpha_{a,t} = \alpha_{p,t} - \alpha_{b,t}$.

The **(ex post) tracking difference** is $e_{t+1} = \alpha'_{a,t} \mathbf{R}_{t+1}$

The important ex-ante quantities are, assuming that we have expected returns μ_t and the covariance matrix Σ_t at the end of period t :

- **expected return (alpha)** relative to benchmark: $\mathbb{E}_t[e_{t+1}] = \alpha'_{a,t} \mu_t$
- **(ex-ante) tracking error**: $\sigma_t[e_{t+1}] = \sqrt{\alpha'_{a,t} \Sigma_t \alpha_{a,t}}$
- **information ratio**: $IR_{p,t} = \frac{\mathbb{E}_t[e_{t+1}]}{\sigma_t[e_{t+1}]} = \frac{\alpha'_{a,t} \mu_t}{\sqrt{\alpha'_{a,t} \Sigma_t \alpha_{a,t}}}$

The *IR* is the equivalent to the *SR* in presence of a benchmark.

Ex-post Performance Relative to a Benchmark

At the end of the investment strategy over T periods, we observe the (ex post) portfolio returns $R_{p,t+1} = \alpha'_{p,t} \mathbf{R}_{t+1}$

- **ex-post tracking difference:** $e_{t+1} = R_{p,t+1} - R_{b,t+1} = \alpha'_{a,t} \mathbf{R}_{t+1}$
- **ex-post Alpha:** $\bar{R}_e = \frac{1}{T} \sum_{t=1}^T e_{t+1}$
- **ex-post tracking error:** $\sigma_e = \sqrt{\frac{1}{T} \sum_{t=1}^T e_t^2}$
- **ex-post information ratio:** $IR_p = \frac{\bar{R}_e}{\sigma_e}$

Another measure of interest to assess the characteristics of a portfolio relative to its benchmark is the **Active Share**:

$$AS_p = \frac{1}{2} \sum_{i=1}^N |\alpha_{p,i,t} - \alpha_{b,i,t}|.$$

We divide by 2 to avoid the redundancy.

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- Estimating the Inputs
- Financial performance
- **Issues with Missing Values**
- Design of an Investment Strategy
- Risk Measures

Issues with Missing Values

A recurrent problem is missing values.

This situation happens in three cases:

- Case 1: The firm does not exist (or was not listed) at the beginning of the sample
- Case 2: The firm is delisted before the end of the sample
- Case 3: There is a missing price at some point in the sample (misreporting, such as a negative price, etc.)

Missing values represent an issue because they will generate missing values in statistics such as the expected return (sample average) or the covariance matrix.

Sample average

In cases 1 and 2, we just compute the average over the available sample.

In case 3, we proxy the missing price by the previous price to avoid look-ahead bias.

Issues with Missing Values – Covariance matrix

Missing values will cause rows and columns of the covariance matrix to be missing values. Any computation based on the covariance matrix (for instance, the plug-in estimator) will fail.

Take two return series, $(R_{i,\tau_{i,1}}, \dots, R_{i,\tau_{i,n_i}})'$ and $(R_{j,\tau_{j,1}}, \dots, R_{j,\tau_{j,n_j}})'$

where

- $\tau_{i,1}$ and τ_{i,n_i} are the first and last dates available for series i
- $\tau_{j,1}$ and τ_{j,n_j} are the first and last dates available for series j

Then, for entries (i, j) and (j, i) of the covariance matrix, we compute the covariance based on the common sample.

Let

$$\tau_1 = \max(\tau_{i,1}, \tau_{j,1}) \quad \text{and} \quad \tau_n = \min(\tau_{i,n_i}, \tau_{j,n_j})$$

be the first and last dates available for both series.

Finally, we have

$$\text{Var}(R_i) = \frac{1}{\tau_{i,n_i} - \tau_{i,1} - 1} \sum_{k=\tau_{i,1}}^{\tau_{i,n_i}} (R_{i,k} - \bar{R}_i)^2$$

$$\text{Var}(R_j) = \frac{1}{\tau_{j,n_j} - \tau_{j,1} - 1} \sum_{k=\tau_{j,1}}^{\tau_{j,n_j}} (R_{j,k} - \bar{R}_j)^2$$

$$\rho(R_i, R_j) = \frac{\frac{1}{\tau_n - \tau_1 - 2} \sum_{k=\tau_1}^{\tau_n} (R_{i,k} - \bar{R}_i)(R_{j,k} - \bar{R}_j)}{\frac{1}{\tau_n - \tau_1 - 1} \sqrt{\left(\sum_{k=\tau_1}^{\tau_n} (R_{i,k} - \bar{R}_i)^2 \right) \left(\sum_{k=\tau_1}^{\tau_n} (R_{j,k} - \bar{R}_j)^2 \right)}}.$$

and

$$\text{Cov}(R_i, R_j) = \rho(R_i, R_j) \times \sqrt{\text{Var}(R_i) \text{Var}(R_j)}.$$

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- Risk Measures

All the strategies we have discussed in Lecture 1 had closed-form solutions.

In practice, as we illustrated previously, optimal weights are in general too extreme for most investors to invest in such strategies.

One obvious limitation is that optimal weights can be very negative, which would not be acceptable for institutional investors.

Consequently, even diversifying strategies will impose restrictions, with positivity of weights in the first place. Because of this (set of) restrictions, the solution does not have a closed form.

We now illustrate the impact of positivity constraints on the optimal solution in the case of the global minimum variance strategy.

Long-Short Minimum-Variance Strategy

The optimal weights of the long-short (i.e., unconstrained) Global Minimum Variance portfolio (GMV) are given by

$$\begin{aligned}\alpha_{\text{GMV}} &= \arg \min_{\{\alpha\}} \alpha' \Sigma \alpha \\ \text{subject to } &\alpha' \mathbf{e} = 1,\end{aligned}$$

where $\mathbf{e}$ denotes the vector of N ones.

The optimal solution is

$$\alpha_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{e}}{\mathbf{e}' \Sigma^{-1} \mathbf{e}}.$$

Remark: This closed-form solution requires the covariance matrix to be invertible.

Long-Only Minimum-Variance Strategy

Now, we consider the long-only (i.e., constrained) Global Minimum Variance portfolio (LMV) problem:

$$\alpha_{\text{LMV}} = \arg \min_{\{\alpha\}} \alpha' \Sigma \alpha$$

subject to

$$\begin{cases} \alpha' \mathbf{e} = 1, \\ \alpha_i \geq 0 \end{cases} \quad \text{for all } i = 1, \dots, N.$$

The resolution requires (among others) to write down the Kuhn–Tucker first-order conditions.

There is no longer a closed-form solution for α_i .

Python Example (GMV Portfolio)

```
# Import and rename library
import scipy.optimize as sco

# Download data (daily returns, etc) -> Define returns : rets

# Variable declaration
number_securities = 4

# Objective function
# (annualized portfolio return based on daily data)
def port_vol(weights):
    return np.sqrt(np.dot(weights.T, np.dot(rets.cov(), weights)))
```

Python Example (GMV Portfolio) (Continued)

```
# Determination of constraints
# 1. Equality constraints
cons = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1})

# Initial conditions/weights for the optimization
init_weights = np.array(number_securities*[1./number_securities, ])

# Output : array([0.25, 0.25, 0.25, 0.25])

# Constrained minimization problem with SLSQP (Sequential Least
    Squares Programming)
opt_ls_gmv = sco.minimize(port_vol, initial_weights, method='SLSQP',
    constraints=cons)
```

Python Example (GMV Portfolio) (Continued)

```
opt_ls_gmv
```

```
# Output :
```

```
message : Optimization terminated successfully
```

```
success : True
```

```
status :0
```

```
fun :0.1094189647332854
```

```
nit :7
```

```
jac :[1.101e^-01 1.088e^-01 1.095e^-01 1.094e^-01]
```

```
nfev :36
```

Python Example (Long-Only Portfolio) (Continued)

```
# Determination of constraints
# 1. Equality constraints
cons = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1})

# 2. Positivity constraints
bnds = tuple((0, 1) for x in range(number_securities))

# Initial conditions/weights for the optimization
init_weights = np.array(number_securities*[1./number_securities, ])

# Output : array([0.25, 0.25, 0.25, 0.25])

# Constrained minimization problem
opt_l_lmv = sco.minimize(port_vol, initial_weights, method='SLSQP',
bounds=bnds, constraints=cons)
```

Python Example (Long-Only Portfolio) (Continued)

```
opt_l_lmv
```

```
#Output :
```

```
message : Optimization terminated successfully
```

```
success :True
```

```
status :0
```

```
fun :0.10942155263459781
```

```
x : [ 0 0.00106171 0.54326307 0.45567522 ]
```

```
nit :9
```

```
jac : [0.11098003, 0.10948556, 0.10939826, 0.10944918]
```

```
nfev :45
```

Python Example (Long-Only Portfolio) (Continued)

```
opt_l_lmv['x'].round(3)
```

```
#Output : array([0, 0.001, 0.543, 0.456])
```

```
port_vol(opt_l_lmv['x']).round(3)
```

```
#Output : 0.109
```

```
port_ret(opt_l_lmv['x']).round(3)
```

```
#Output : 0.06
```

```
port_ret(opt_l_lmv['x'])/port_vol(opt_l_lmv['x'])
```

```
#Output : 0.5504173642829329
```

Note : `port_ret` is a standard function to determine the (annualized) portfolio return (based on daily data)

```
def port_ret(weights) :
```

```
    return np.sum(rets.mean() * weights)
```

Out-of-sample Analysis: Inputs

“Out of sample”: we use only past data to compute the optimal portfolio for the next period. Assume that we rebalance our portfolio each end of year (31st of December).

Example: we have monthly returns starting in January 2000 to December 2025. We use the first 10 years of monthly returns (from Jan. 2004 to Dec. 2013) to compute the vector of expected returns and the covariance matrix.

Therefore, we have $\tau = 10 \times 12 = 120$ months of data available to compute the expected returns and the covariance matrix valid for the allocation at the end of Dec. 2013 for the allocation for Dec. 2014.

Dec. 2013 corresponds to year $Y_0 = 2013$ for annual data and to $t_0 = \tau = 120$ for monthly data. Dec. 2014 corresponds to $Y_0 + 1 = 2014$ and $t_0 + 12 = 132$.

We associate year Y to month t (Dec. of the year).

Out-of-sample Analysis: Inputs

At the end of year Y , we compute the expected returns for year $Y + 1$ as

$$\hat{\mu}_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} \mathbf{R}_{t-k}.$$

Similarly, the covariance matrix is

$$\Sigma_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} (\mathbf{R}_{t-k} - \hat{\mu}_Y)(\mathbf{R}_{t-k} - \hat{\mu}_Y)'$$

The allocation is determined at the end of year Y for year $Y + 1$.

We compute the optimal allocation of the portfolio using Markowitz's approach with a risk aversion parameter λ .

Out-of-sample Analysis: Weights

In general, the covariance matrix Σ_Y is not invertible, so using the closed-form solution for the optimal weights will not work. If we impose positivity of weights, the closed-form solution does not apply anyway.

Instead, we use the following minimization problem, while restricting the optimal weights to be positive. For instance, for the long-only minimum variance portfolio, we choose the weights at the end of year Y as follows:

$$\alpha_{\text{LMV},Y} = \arg \min_{\{\alpha_Y\}} \alpha_Y' \Sigma_Y \alpha_Y$$

subject to

$$\begin{cases} \alpha_Y' \mathbf{e} = 1, \\ \alpha_{Y,i} \geq 0 \end{cases} \quad \text{for all } i = 1, \dots, N.$$

Then, we roll the window by one year and iterate until the end of the sample, so that our portfolio is rebalanced every year from Dec. 2013 to Dec. 2024.

Out-of-sample Analysis: Ex-post Performance

Finally, we compute the ex-post performance of the portfolio.

The portfolio return is computed every month of year $Y + 1$ using monthly stock returns of year $Y + 1$ and optimal weights calculated at the end of year Y . This gives:

$$R_{p,t+k} = \alpha'_{t+k-1} \mathbf{R}_{t+k} \quad \text{for } k = 1, \dots, 12,$$

where α_{t+k-1} is the vector of weights in month $k - 1$ of year $Y + 1$ (we use $\alpha_t = \alpha_Y$ for January return $R_{p,t+1}$). This weight accounts for the relative change in the value of the securities in the portfolio:

$$\alpha_{i,t+k-1} = \alpha_{i,t+k-2} \times \frac{1 + R_{i,t+k-1}}{1 + \mathbf{R}_{p,t+k-1}} \quad \text{for } k = 2, \dots, 12.$$

For instance, for the return at the end of March ($R_{p,t+3}$), we compute the weight at the end of February (α_{t+2}) by updating the weights of January (α_{t+1}) for the relative performance of securities in February ($R_{p,t+2}$) ($k = 3$).

Out-of-sample Analysis: Ex-post Performance (timeline)

Date	Portfolio return	Weights update
Dec. Y		$\alpha_{i,t} = \alpha_{i,Y}$ (optimal weight)
Jan. $Y+1$	$R_{p,t+1} = \alpha'_t \mathbf{R}_{t+1}$	$\alpha_{i,t+1} = \alpha_{i,t} \times \frac{1 + R_{i,t+1}}{1 + R_{p,t+1}}$
Feb. $Y+1$	$R_{p,t+2} = \alpha'_{t+1} \mathbf{R}_{t+2}$	$\alpha_{i,t+2} = \alpha_{i,t+1} \times \frac{1 + R_{i,t+2}}{1 + R_{p,t+2}}$
$\vdots$	$\vdots$	$\vdots$
Dec. $Y+1$	$R_{p,t+12} = \alpha'_{t+11} \mathbf{R}_{t+12}$	$\alpha_{i,t+12} = \alpha_{i,t+11} \times \frac{1 + R_{i,t+12}}{1 + R_{p,t+12}}$

We now have a time series of ex post portfolio returns: $\{R_{p,\tau+1}, \dots, R_{p,T}\}$, where T is the total number of months in the sample.

Out-of-sample Analysis: Ex-post Performance measures

From the ex-post return series, we can compute ex-post performance measures:

- **Ex-post average return:**

$$\bar{R}_p = \frac{1}{T - \tau} \sum_{t=\tau+1}^T R_{p,t}.$$

- **Ex-post volatility:**

$$\sigma_p = \sqrt{\frac{1}{T - \tau} \sum_{t=\tau+1}^T (R_{p,t} - \bar{R}_p)^2}.$$

- **Ex-post Sharpe ratio:**

$$SR_p = \frac{\bar{R}_p - R_f}{\sigma_p}.$$

Equal-Weighted and Value-Weighted Portfolios

Some portfolios of interest do not require any particular computation.

The weights of the **equal-weighted portfolio** are $\alpha_{i,t}^{(ew)} = 1/N$ if there are N securities in the portfolio. The portfolio return is just the simple average of securities returns:

$$R_{t+1}^{(ew)} = \frac{1}{N} \sum_{i=1}^N R_{i,t+1}.$$

The weights of the **value-weighted portfolio** are $\alpha_{i,t}^{(vw)} = \frac{Cap_{i,t}}{\sum_{j=1}^N Cap_{j,t}}$ where $Cap_{i,t}$ is the market capitalization **at the end of period t** . The portfolio return is given by

$$R_{t+1}^{(vw)} = \sum_{i=1}^N \alpha_{i,t}^{(vw)} R_{i,t+1}.$$

Remark: You must use the weights from the previous period ($\alpha_{i,t}^{(vw)}$ for computing $R_{t+1}^{(vw)}$).

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Loss of a Portfolio

Portfolio: collection of stocks or bonds, book of derivatives, collection of risk loans, financial institution's overall position in risky assets.

Value of the portfolio at time t : V_t (Observable at time t)

Loss of the portfolio over the period $[t, t + \Delta]$:

$$L_{t,t+\Delta} = -(V_{t+\Delta} - V_t)$$

$L_{t,t+\Delta}$ is observable at time $t + \Delta$, but not at time t .

Remark 1: In these notations, the loss is a positive number. Remark 2: In general, Δ is a fixed horizon, like 1 day. It gives $L_{t+1} = -(V_{t+1} - V_t)$.

Example: Stock portfolio

We consider a **portfolio of N stocks**. The price process of stock i is $\{P_{i,t}\}_t$. The investor has bought $N_{i,t}$ shares of stock i .

The value at $t + 1$ of the portfolio is:

$$V_{t+1} = \sum_{i=1}^N N_{i,t} P_{i,t+1} = V_t \sum_{i=1}^N \left(\frac{N_{i,t} P_{i,t}}{V_t} \frac{P_{i,t+1}}{P_{i,t}} \right) = V_t \sum_{i=1}^N \alpha_{i,t} (1 + R_{i,t+1}),$$

where $\alpha_{i,t} = N_{i,t} P_{i,t} / V_t$ is the relative weight of stock i at time t .

The loss is:

$$L_{t+1} = -(V_{t+1} - V_t) = -V_t \sum_{i=1}^N \alpha_{i,t} R_{i,t+1}.$$

Given the mean and covariance matrix of the distribution of stock returns, it is easy to compute the first and second moments of the distribution of the loss L_{t+1} .

Risk Measures based on Loss distribution

There are **two major problems with loss distributions**:

- Any estimate of the loss distribution is **based on past data**. It is not clear whether the past is really useful to predict the future.
- Even in a stationary environment, it is difficult to estimate the loss distribution accurately. It will be subject to **estimation error**.

We consider a portfolio of risky assets and a fixed time horizon Δ . We denote by $F_L(l) = \mathbb{P}[L \leq l]$ the cdf of the loss distribution.

If the loss distribution is time varying, we use the notation:

$$F_{L,t}(\ell) = \Pr[L_{t+1} \leq \ell \mid \mathcal{I}_t].$$

Remark: The distribution of returns is just the distribution of $(V_{t+1} - V_t)/V_t = -L_{t+1}/V_t$. We can work with one or the other.

Definition: The VaR of a portfolio is the minimum potential loss that the portfolio can suffer in the $(1 - \theta)\%$ worst cases, over a given time horizon Δ .

Continuous distribution:

$$\Pr[L_{t+1}(\alpha_t) > \text{VaR}_{\theta,t} | \mathcal{I}_t] = 1 - \theta \quad \text{or} \quad \text{VaR}_{\theta,t} = F_{L,t}^{-1}(\theta),$$

where $L_{t+1}(\alpha_t) = -(V_{t+1}(\alpha_t) - V_t(\alpha_t))$ is the loss of the portfolio between t and $t + 1$, and $\mathcal{I}_t$ is the information set at date t .

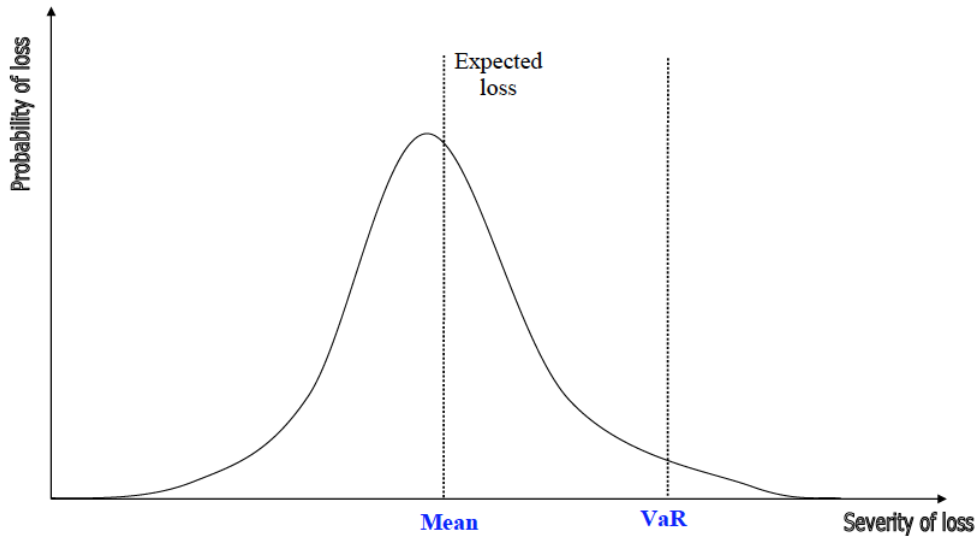
Non-continuous distribution:

$$\text{VaR}_{\theta,t} = \inf \left\{ \ell \in \mathbb{R} \mid \Pr[L_{t+1}(\alpha_t) \leq \ell | \mathcal{I}_t] > \theta \right\}.$$

Remarks:

- The VaR is simply the θ -quantile of the conditional loss cdf (take $\theta = 99\%$).
- The VaR does not give any information about the severity of losses that occur with a probability less than $1 - \theta$.

Loss distribution and Value-at-Risk



Value-at-Risk (VaR): Return-based definition

Alternatively, the VaR can be defined as a quantile of the return distribution:

$$\Pr\left[\frac{L_{t+1}(\alpha_t)}{V_t(\alpha_t)} > \frac{\text{VaR}_{\theta,t}}{V_t(\alpha_t)} \mid \mathcal{I}_t\right] = 1 - \theta \quad \text{or} \quad \Pr[R_{t+1}(\alpha_t) \leq -\overline{\text{VaR}}_{\theta,t} \mid \mathcal{I}_t] = 1 - \theta$$

where $\overline{\text{VaR}}_{\theta,t} = \text{VaR}_{\theta,t}/V_t(\alpha_t)$ denotes the VaR for probability θ for 1\$ invested.

If we define the conditional cdf $F_{R,t}(x) = \Pr[R_{p,t+1} \leq x \mid \mathcal{I}_t]$, we also have

$$\overline{\text{VaR}}_{\theta,t} = -F_{R,t}^{-1}(1 - \theta).$$

The VaR is simply (minus) the $(1 - \theta)$ -quantile of the conditional cdf of the portfolio return.

If returns were i.i.d., then the VaR would be constant over time. It would be simply given by the inverse of the unconditional cdf:

$$\overline{\text{VaR}}_{\theta} = -F_R^{-1}(1 - \theta),$$

where $F_R(x) = \Pr[R_{p,t+1} \leq x]$ is the cdf of the portfolio return.

Practical parameter choice

- Market risk: $\theta = 0.99$ and $\Delta = 10$ days
- Trading desk limit: $\theta = 0.95$ and $\Delta = 1$ day

Caveats:

- VaR does not describe the worst loss.
- VaR does not describe the losses in the left tail.
- VaR is subject to estimation error (**model risk**).
- VaR neglects problems related to **market liquidity**. In crash periods, it may be impossible to liquidate a position. One should take the cost of liquidation into account.

Expected Shortfall (ES)

Definition: The Expected Shortfall ES is the expected value of the loss of the portfolio in the $(1 - \theta)\%$ worst cases, over a given time horizon Δ .

Continuous distribution:

$$\text{ES}_{\theta,t} = \mathbb{E}[L_{t+1} \mid L_{t+1} \geq \text{VaR}_{\theta,t}].$$

Non-continuous distribution:

$$\text{ES}_{\theta,t} = \mathbb{E}[L_{t+1} \mid L_{t+1} > \text{VaR}_{\theta,t}] \frac{\Pr[L_{t+1} > \text{VaR}_{\theta,t}]}{1 - \theta} + \text{VaR}_{\theta,t} \frac{(1 - \theta) - \Pr[L_{t+1} > \text{VaR}_{\theta,t}]}{1 - \theta}.$$

- ES depends only on the distribution of L_{t+1} . We have $\text{ES}_{\theta,t} \geq \text{VaR}_{\theta,t}$.
- ES is also the average VaR over all levels $u \geq \theta$.

Alternatively, the Expected Shortfall can be written for the return distribution:

$$\overline{\text{ES}}_{\theta,t} = -\mathbb{E}\left[R_{p,t+1} \mid R_{p,t+1} \leq F_{R,t}^{-1}(1 - \theta)\right],$$

where $\overline{\text{ES}}_{\theta,t}$ denotes the ES for probability θ for 1\$ invested.

Expected Shortfall: Properties

Main properties of the ES:

- ES is a coherent measure of risk. It can be reduced by diversification.
- ES directly controls the risk in the left tail of the distribution.

ES is the average of the $(1 - \theta)T$ highest losses (T = number of observations). Therefore,

$$\begin{aligned} \text{ES}_\theta(X) + \text{ES}_\theta(Y) &= [\text{avg of } (1 - \theta)T \text{ highest losses of } X] \\ &\quad + [\text{avg of } (1 - \theta)T \text{ highest losses of } Y] \\ &\geq [\text{avg of } (1 - \theta)T \text{ highest losses of } (X + Y)] = \text{ES}_\theta(X + Y). \end{aligned}$$

The only case where we can have $\text{ES}_\theta(X) + \text{ES}_\theta(Y) = \text{ES}_\theta(X + Y)$ is when the worst X and Y occur in the same events.

Definition: Coherent measure of risk

Artzner, Delbaen, Eber, and Heath (1999) proposed a set of conditions for coherency.

Definition:

Let V be a set of real-valued random variables (typically, the final loss of the portfolio). The function $\rho : V \rightarrow \mathbb{R}$ is a coherent risk measure if it satisfies:

- 1 **Translation invariance:** $L \in V$, $c \in \mathbb{R}$, then $\rho(L + c) = \rho(L) - c$

Adding or subtracting a deterministic (safe) amount c to a position with loss L will alter the risk measure by exactly the same amount c .

- 2 **Subadditivity:** $L_1, L_2 \in V$, then $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$.

The risk of a portfolio constituted of two sub-portfolios is smaller than the sum of the risks of the two sub-portfolios.

- 3 **Positive homogeneity:** $L \in V$, $\lambda \geq 0$, then $\rho(\lambda L) = \lambda \rho(L)$

Increasing the size of the portfolio by a factor λ increases the risk measure by the same factor λ .

- 4 **Monotonicity:** $L_1, L_2 \in V$, with $L_1 \leq L_2$, then $\rho(L_1) \leq \rho(L_2)$ almost surely

Positions that lead to higher losses in every state of the world require more risk capital.

Definition: Coherent measure of risk

Subadditivity raises several issues:

- Using a non-subadditive risk measure in a Markowitz' problem may lead to very concentrated optimal portfolio.
- If regulators use non-subadditive risk measures in determining the regulatory capital for financial institutions, the institutions may have an incentive to legally break up into various subsidiaries in order to reduce its regulatory capital requirements.
- Subadditivity makes decentralization of risk management possible. Consider two trading desks with associated losses L_1 and L_2 . If we want to make sure that the overall loss $L = L_1 + L_2$ is smaller than some number M , we just have to choose bounds M_1 and M_2 such that $M_1 + M_2 = M$, and impose on each of the desks the constraint that $L_1 \leq M_1$ and $L_2 \leq M_2$.

Coherence properties: subadditivity and super-additivity

The **variance** is subadditive when the covariance is negative:

$$\text{Var}(L_1 + L_2) = \text{Var}[L_1] + \text{Var}[L_2] + 2\rho_{L_1, L_2} \sqrt{\text{Var}[L_1]\text{Var}[L_2]} \leq \text{Var}[L_1] + \text{Var}[L_2]$$

and super-additive when the covariance is positive:

$$\text{Var}(L_1 + L_2) = \text{Var}[L_1] + \text{Var}[L_2] + 2\rho_{L_1, L_2} \sqrt{\text{Var}[L_1]\text{Var}[L_2]} \geq \text{Var}[L_1] + \text{Var}[L_2]$$

The **standard deviation** is always subadditive:

$$\begin{aligned} \text{Var}(L_1 + L_2) &= \text{Var}[L_1] + \text{Var}[L_2] + 2\rho_{L_1, L_2} \sqrt{\text{Var}[L_1]\text{Var}[L_2]} \\ &\leq \text{Var}[L_1] + \text{Var}[L_2] + 2\sqrt{\text{Var}[L_1]\text{Var}[L_2]} = \left(\sqrt{\text{Var}[L_1]} + \sqrt{\text{Var}[L_2]}\right)^2 \end{aligned}$$

so that

$$\sqrt{\text{Var}[L_1 + L_2]} \leq \sqrt{\text{Var}[L_1]} + \sqrt{\text{Var}[L_2]}$$

Variance and standard deviation: Positive Homogeneity

The **variance and standard deviation** fail translation invariance:

$$\text{Var}[L + c] = \text{Var}[L] \quad \text{and} \quad \sqrt{\text{Var}[L + c]} = \sqrt{\text{Var}[L]}$$

The **variance** is not positive homogeneous:

$$\text{Var}[\lambda L] = \lambda^2 \text{Var}[L]$$

The **standard deviation** is positive homogeneous:

$$\sqrt{\text{Var}[\lambda L]} = \lambda \sqrt{\text{Var}[L]}$$

VaR is not a Coherent Measure of Risk

The VaR of a portfolio is the minimum potential loss that the portfolio can suffer in the $(1 - \theta)\%$ worst cases, over a given time horizon Δ :

$$\text{VaR}_{\theta,t} = \inf \left\{ \ell \in \mathbb{R} \mid \Pr[L_{t+1}(\alpha_t) \leq \ell \mid \mathcal{I}_t] > \theta \right\}.$$

The VaR does not satisfy the subadditivity condition, so that diversification does not necessarily result in a reduction of the VaR.

Example: We have two identical bonds, A and B . Each bond defaults with probability 4%. We get a loss of 100 if default occurs, and a loss of 0 if no default occurs. The 95% VaR of each bond is therefore 0, so that

$$\text{VaR}_{95\%,t}(A) + \text{VaR}_{95\%,t}(B) = 0.$$

VaR is not a Coherent Measure of Risk (Cont'd)

Now suppose that defaults are independent. Elementary calculations then establish that we get:

- a loss of 0 with probability $(0.96)^2 = 0.9216$,
- a loss of 200 with probability $(0.04)^2 = 0.0016$,
- a loss of 100 with probability $1 - 0.9216 - 0.0016 = 0.0768$.

So,

$$\text{VaR}_{95\%,t}[A + B] = 100 > \text{VaR}_{95\%,t}[A] + \text{VaR}_{95\%,t}[B] = 0 \quad \Rightarrow \quad \text{no subadditivity.}$$

Remark: VaR can be made subadditive by imposing some restrictions on the loss.

Artzner et al. (1999) propose the Expected Shortfall (ES) as an alternative risk measure.

Expected Shortfall is a Coherent Measure of Risk

The Expected Shortfall ES is the expected value of the loss of the portfolio in the $(1 - \theta)\%$ worst cases over a given time horizon Δ (non-continuous distribution):

$$ES_{\theta,t} = \mathbb{E}[L_{t+1} \mid L_{t+1} > VaR_{\theta,t}] \frac{\Pr[L_{t+1} > VaR_{\theta,t}]}{1 - \theta} + VaR_{\theta,t} \frac{(1 - \theta) - \Pr[L_{t+1} > VaR_{\theta,t}]}{1 - \theta}$$

The expected shortfall of bond A is given by ($VaR_{95\%,t}(A) = 0$):

$$\begin{aligned} ES_{95\%,t}[A] &= \mathbb{E}[L_{t+1} \mid L_{t+1} > 0] \frac{\Pr[L_{t+1} > 0]}{5\%} + VaR_{95\%,t}[A] \frac{5\% - \Pr[L_{t+1} > 0]}{5\%} \\ &= 100 \times \frac{4}{5} = 80 \end{aligned}$$

so that

$$ES_{95\%,t}[A] + ES_{95\%,t}[B] = 160.$$

Expected Shortfall is a Coherent Measure of Risk (Cont'd)

The expected shortfall of portfolio $A + B$ is given by ($\text{VaR}_{95\%,t}(A + B) = 100$):

$$\begin{aligned}\text{ES}_{95\%,t}(A + B) &= \mathbb{E}[L_{t+1} \mid L_{t+1} > 100] \frac{\Pr[L_{t+1} > 100]}{5\%} \\ &\quad + \text{VaR}_{95\%,t}(A + B) \frac{5\% - \Pr[L_{t+1} > 100]}{5\%} \\ &= 200 \times \frac{0.16\%}{5\%} + 100 \times \frac{5\% - 0.16\%}{5\%} = 103.2\end{aligned}$$

So,

$$\text{ES}_{95\%,t}(A + B) < \text{ES}_{95\%,t}(A) + \text{ES}_{95\%,t}(B),$$

which is consistent with subadditivity.

End of Lecture 5